

# Link: A Modular Exoskeleton System Designed for Dynamic Manual Tasks

Sean Bridges, Nikki Esposito, Reese Peterson, Stefan Fasano, Garrett Lundegard, Alexis Maslyczyk, Kain Miller, Bridget Longo, Chinmay Shah, Nicole Stafford, Jared M. Li

## 1. Abstract

### Why Is This Important?

- Repetitive lifting, lowering, and carrying tasks common to materials handling occupations can lead to fatigue, which could cause injury.
- Exoskeletons have been introduced to worksites to assist repetitive motions and reduce injury incidence.
- Most devices lack the adaptability and capability to assist a diverse taskset, hindering large-scale usefulness and adoption.

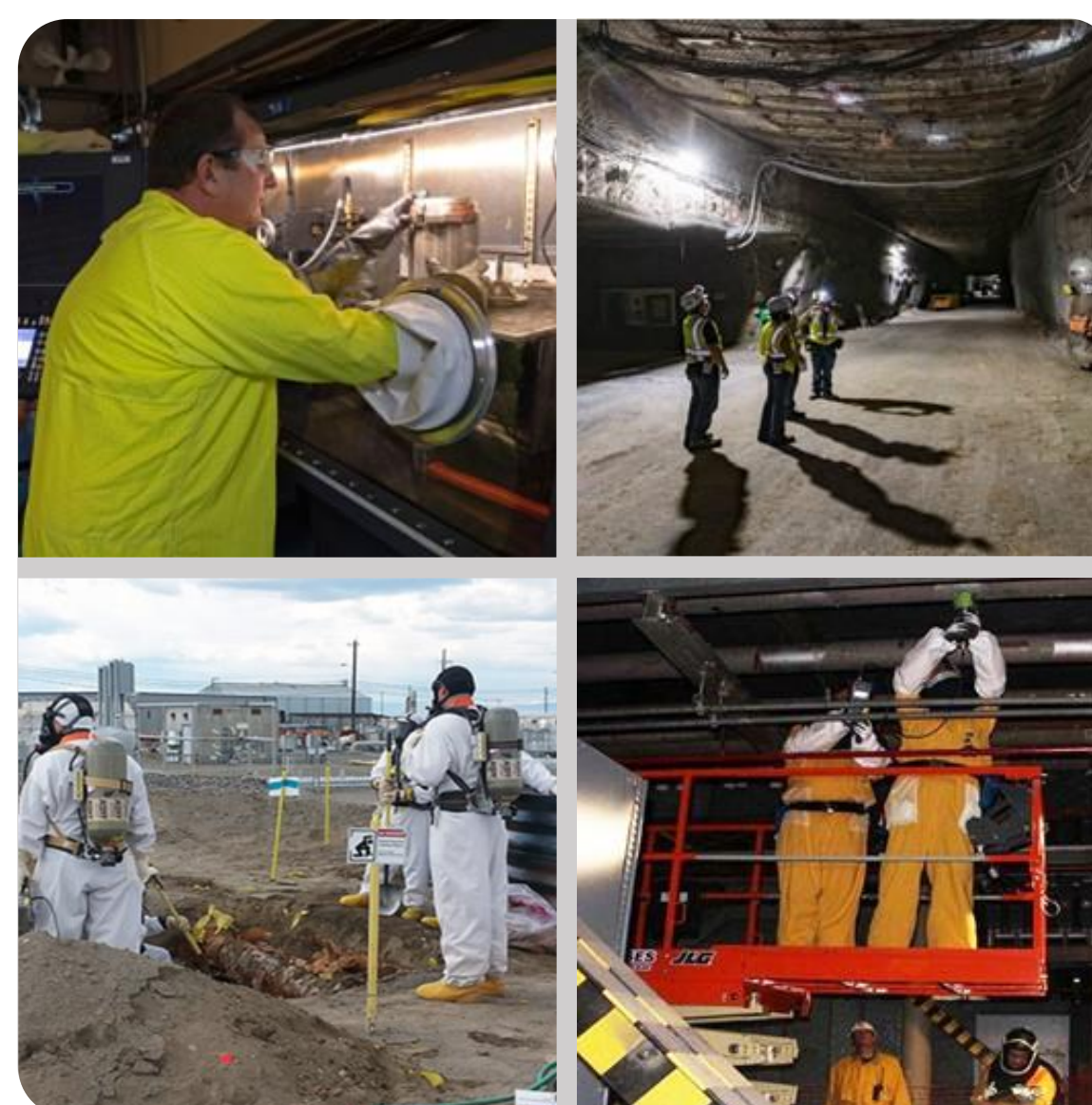


Figure 1: Manual materials handling tasks are diverse and can be fatiguing, which could lead to injury.

### Novel Contribution

- We introduce Link: a custom modular exoskeleton configurable to a wide range of manual tasks featuring:
  - Actuator packages with integrated power electronics
  - Swappable carbon fiber linkages with built-in IMUs
  - Modular linkages and cuffing interfaces, dictated by user anatomy and morphology.
  - Automatic detection and controls adjustment of linkage sizes and joint configurations
  - 68 Nm peak torque, 17 rad/s max velocity per joint
  - 3 hour battery life while running at low-torque

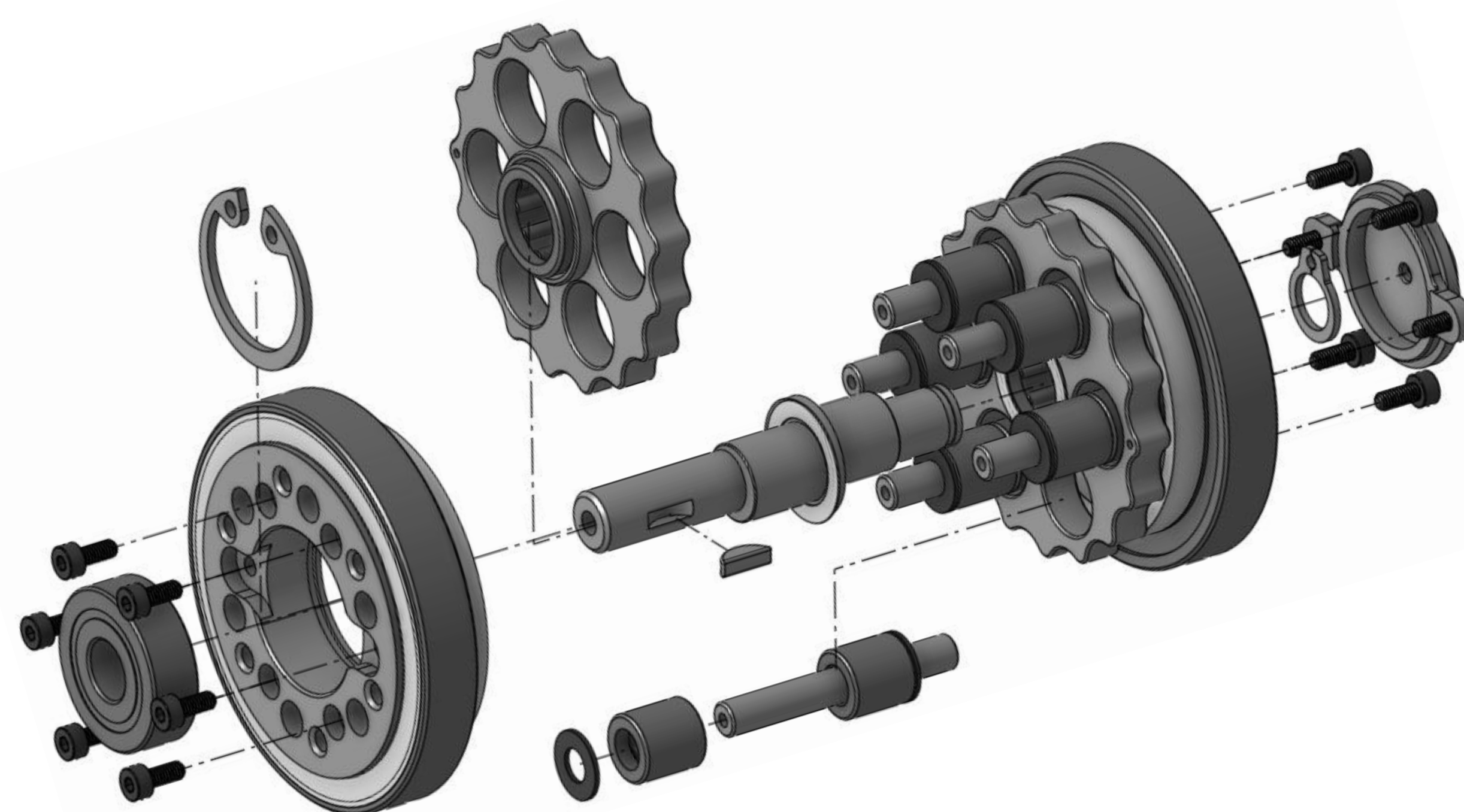


Figure 2: Custom 17:1 cycloidal actuation transmission employing rolling contact bearing surfaces for improved transmission efficiency with minimal viscous and static friction properties

## 2. Exoskeleton Device



Figure 3: Multiple views of the Link exoskeleton, shown in both the hip-only and hip-knee configurations simultaneously. Key features are indicated.

## 3. Control Methodology

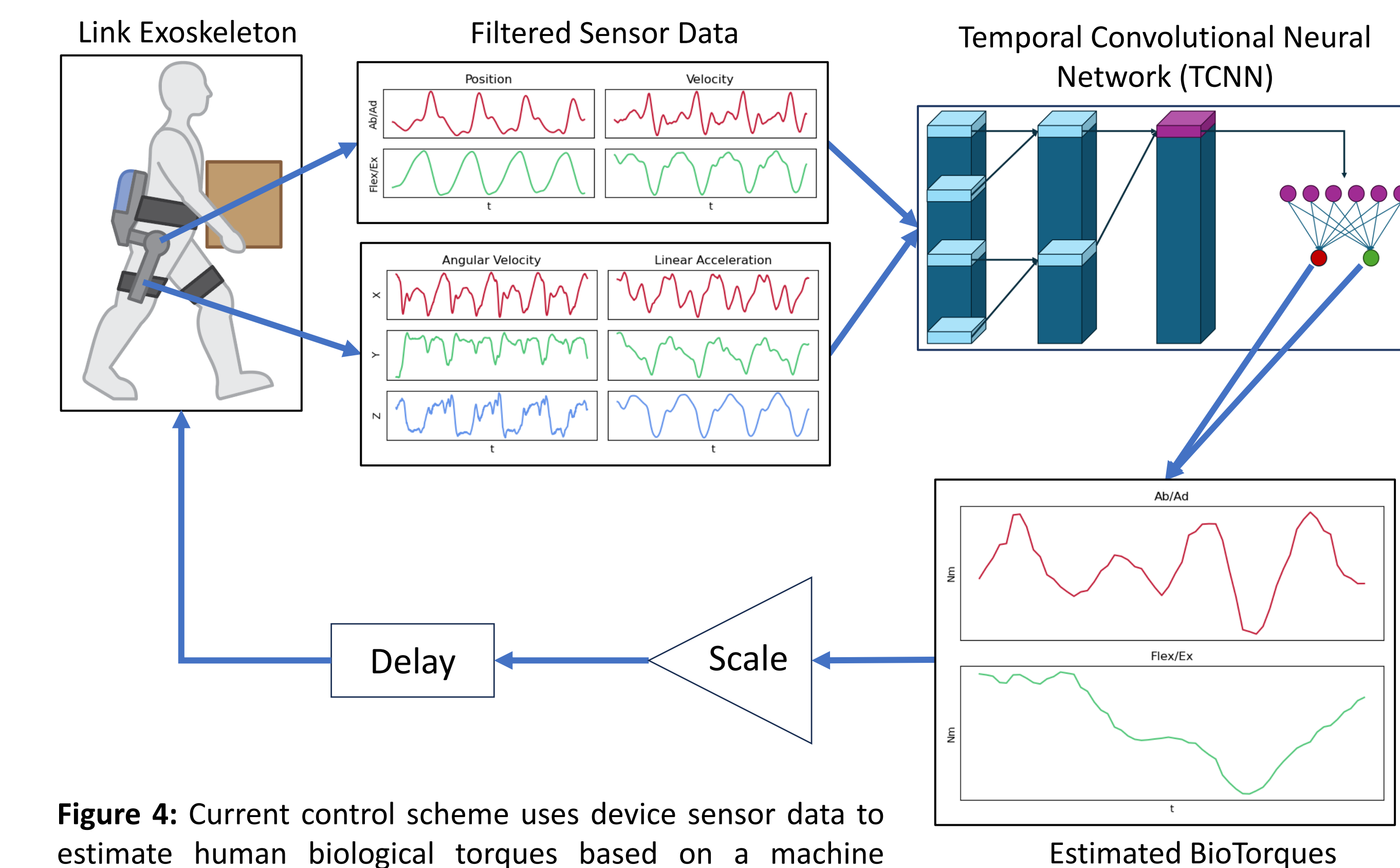


Figure 4: Current control scheme uses device sensor data to estimate human biological torques based on a machine learning model that was trained with biomechanical data.

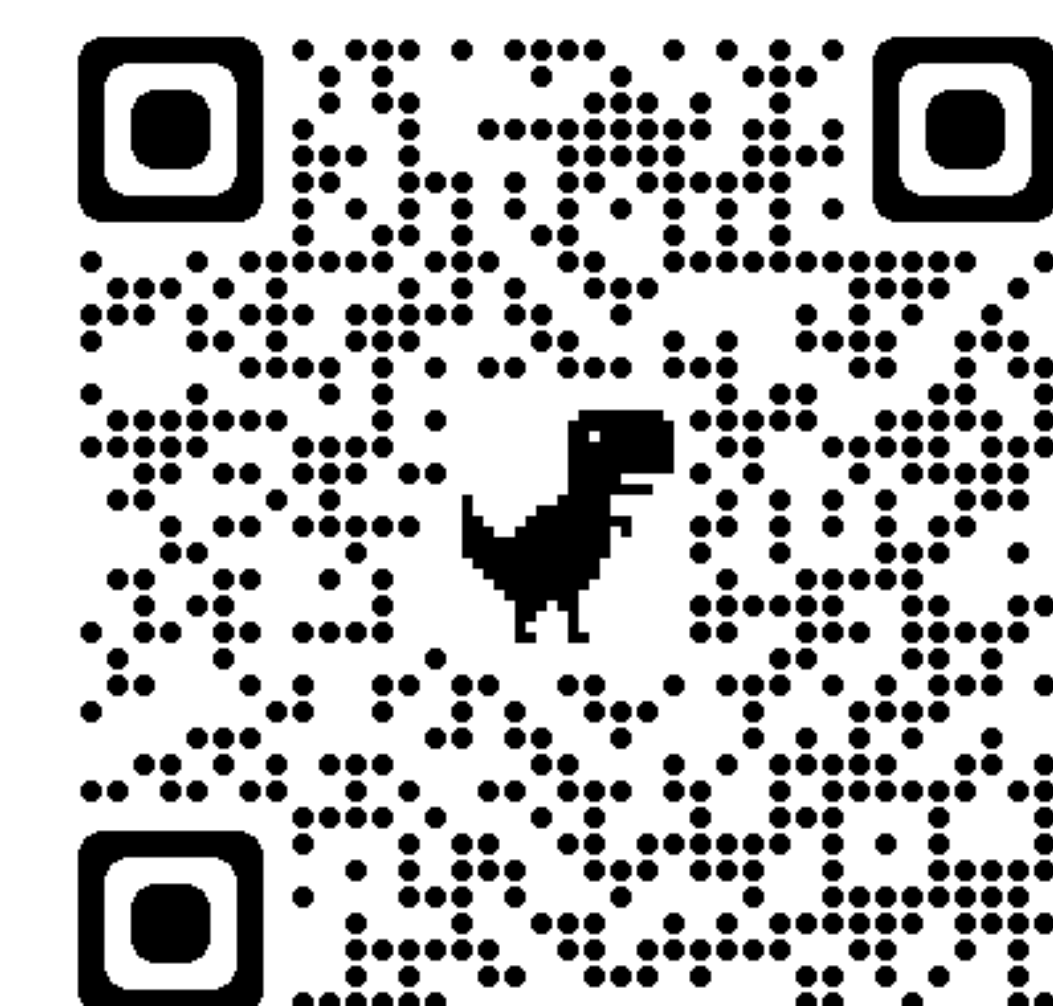
## 4. Future Work

- Run device validation and collect human outcomes.
- Integrate custom sensor suit to improve biomechanical estimation and assistance.
- Explore assistance personalization through optimization methods.
- Develop new Link modules for upper body assistance and for load carriage.



Figure 5: Link's modularity allows for a wide breadth of future use cases in dangerous work that requires dynamic motion.

## Sponsors / Special Thanks



robots.ihmc.us